

🌸 **MERN Stack Internship – Task 2**

📌 **Task Description**

In Task-1, you created your first MERN app and displayed data from an API. Now, let's move one step ahead and build a **To-Do List Application**.

This project will help you understand how the **frontend (React)** talks to the **backend (Express)** and how data is stored in **MongoDB**

You will learn the most common operations in real applications: **Add new data and View existing data** (also called CRUD – Create and Read).

Think of this project as the foundation for any future MERN application.

🔧 **What You'll Build (Requirements)**

1. Frontend (React):

- A text input and “Add Task” button.
- A task list to display all tasks fetched from the backend.

2. Backend (Express + MongoDB):

- Route: POST /add → To add a new task.
- Route: GET /tasks → To display all tasks.
- Store data in MongoDB Atlas (cloud database).

3. Integration:

- React will send data to backend (Add Task).
- Backend will save data in MongoDB.
- React will fetch and display the saved tasks.

Why This Task Is Important

- Teaches you how frontend and backend communicate.
- Introduces API calls (fetching data from a server).
- Gives you experience with MongoDB database integration.

✅ **Deliverable**

A **MERN To-Do List app** where users can add new tasks and see them displayed immediately.

Sample Starter Code (Simplified)

Backend – server.js

js

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");

const app = express();
app.use(cors());
app.use(express.json());

// Connect MongoDB
mongoose.connect("your_mongodb_url")
  .then(() => console.log("MongoDB Connected"));

const Task = mongoose.model("Task", new mongoose.Schema({ text: String }));

// Add new task
app.post("/add", async (req, res) => {
  const newTask = new Task(req.body);
  await newTask.save();
  res.send(newTask);
});

// Get all tasks
app.get("/tasks", async (req, res) => {
  const tasks = await Task.find();
  res.send(tasks);
});

app.listen(5000, () => console.log("Server running on port 5000"));
```

Frontend – App.js

jsx

```
import React, { useEffect, useState } from "react";
import axios from "axios";

function App() {
  const [tasks, setTasks] = useState([]);
  const [text, setText] = useState("");

  // Fetch tasks from backend
  useEffect(() => {
    axios.get("http://localhost:5000/tasks").then(res => setTasks(res.data));
  }, []);

  // Add new task
  const addTask = () => {
    if(!text) return alert("Task cannot be empty!");
    axios.post("http://localhost:5000/add", { text })
      .then(res => setTasks([...tasks, res.data]));
    setText(""); // reset input
  };

  return (
    <div>
      <h1>MERN To-Do App</h1>
      <input value={text} onChange={e => setText(e.target.value)} />
      <button onClick={addTask}>Add</button>
      <ul>{tasks.map((t) => <li key={t._id}>{t.text}</li>)}</ul>
    </div>
  );
}

export default App;
```

Helpful Resources

Free Videos / Courses

- [MERN To-Do App – FreeCodeCamp](#)
- [React JS Full Course – FreeCodeCamp](#)
- [Node.js Crash Course – Traversy Media](#)

References

- [React Docs](#)
- [Express Docs](#)
- [MongoDB Docs](#)

✂ Tools

- MongoDB Atlas → free cloud DB
- Postman → test your APIs
- Visual Studio Code → code editor